



Leibnitz Theorem



Leibnitz Theorem:

If $y = uv$ where u & v are functions of x possessing n^{th} derivative, then

$$\begin{aligned} y_n = (uv)_n &= {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_r u_{n-r} v_r + \dots + \\ & {}^nC_{n-1} u_1 v_{n-1} + {}^nC_n uv_n. \\ &= u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2!} u_{n-2} v_2 + \dots + \frac{n!}{r!(n-r)!} u_{n-r} v_r + \dots + n u_1 v_{n-1} + uv_n. \end{aligned}$$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$



Successive Differentiation:

Successive differentiation is the process of differentiating a given function successively n times & the result of such differentiation are called successive derivatives.

Common notation of higher order derivatives of $y = f(x)$

1st derivative: $f'(x)$ or y' or $\frac{dy}{dx}$ or Dy or y_1 .

2nd derivative: $f''(x)$ or y'' or $\frac{d^2y}{dx^2}$ or D^2y or y_2 .

3rd derivative: $f'''(x)$ or y''' or $\frac{d^3y}{dx^3}$ or D^3y or y_3 .



Calculation of n^{th} derivatives:

$$1) y = e^{ax}$$

$$y_1 = ae^{ax}$$

$$y_2 = a^2e^{ax}$$

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$$y_n = a^n e^{ax}$$



2) $y = (ax+b)^m$; m is a positive integer greater than n .

$$y_1 = ma(ax+b)^{m-1}$$

$$y_2 = m(m-1)a^2(ax+b)^{m-2}$$

$$y_3 = m(m-1)(m-2)a^3(ax+b)^{m-3}$$

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$$y_n = m(m-1)\dots(m-(n-1))a^n(ax+b)^{m-n}$$

$$= m(m-1)\dots(m-n+1)an(ax+b)^{m-n}$$



$$3)y = \sin(ax+b)$$

$$y_1 = a\cos(ax+b) = a\sin(ax+b+\frac{\pi}{2})$$

$$y_2 = a^2\cos(ax+b+\frac{\pi}{2}) = a^2\sin(ax+b+\frac{2\pi}{2})$$

$$y_3 = a^3\cos(ax+b+\frac{2\pi}{2}) = a^3\sin(ax+b+\frac{3\pi}{2})$$

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$$y_n = a^n\sin(ax+b+\frac{n\pi}{2})$$



4) $y = \cos(ax+b)$.

$$y_1 = -a \sin(ax+b) = a \cos(ax+b+\frac{\pi}{2})$$

$$y_2 = -a^2 \sin(ax+b+\frac{\pi}{2}) = a^2 \cos(ax+b+\frac{2\pi}{2})$$

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$$y_n = a^n \cos(ax+b+\frac{n\pi}{2})$$



$$\begin{aligned} 5) y &= \frac{1}{(x+1)(x+2)} \\ &= \frac{A}{(x+1)} + \frac{B}{(x+2)} = \frac{1}{x+1} - \frac{1}{x+2} \end{aligned}$$

$$\text{Let } u = \frac{1}{x+1} \text{ \& \; } v = \frac{1}{x+2}$$

$$u = (x+1)^{-1}$$

$$u_1 = (-1)(x+1)^{-2}$$

$$u_2 = (-1)(-2)(x+1)^{-3} = 2!(x+1)^{-3}$$

$$u_3 = (-1)(-2)(-3)(x+1)^{-4} = -3!(x+1)^{-4}$$

$$u_n = (-1)^n n! \frac{1}{(x+1)^{n+1}}$$

$$\text{Similarly we can show that } v_n = (-1)^n n! \frac{1}{(x+2)^{n+1}}$$

$$\text{hence } y_n = (-1)^n n! \frac{1}{(x+1)^{n+1}} - (-1)^n n! \frac{1}{(x+2)^{n+1}}$$



6) $y = \frac{1}{(x+2)(x+3)}$. Find y_n .

$$y_n = (-1)^n n! \frac{1}{(x+2)^{n+1}} - (-1)^n n! \frac{1}{(x+3)^{n+1}}$$

7) $y = \cos x \cos 2x$.

$$= \frac{1}{2} [\cos 3x + \cos x]$$

$$\text{Let } u = \frac{1}{2} \cos 3x \quad v = \frac{1}{2} \cos x$$

$$u_n = \frac{3^n}{2} \cos \left(3x + \frac{n\pi}{2} \right) \quad v_n = \frac{1}{2} \cos \left(x + \frac{n\pi}{2} \right)$$

$$\{\text{If } y = \cos(ax+b) \text{ then } y_n = a^n \cos(ax+b+\frac{n\pi}{2})\}$$

$$y_n = \frac{3^n}{2} \cos \left(3x + \frac{n\pi}{2} \right) + \frac{1}{2} \cos \left(x + \frac{n\pi}{2} \right)$$



$$8) y = \frac{1}{2}(\cos 3x + \cos x)$$

$$y_n = \frac{1}{2} \left[3^n \cos\left(\frac{n\pi}{2} + 3x\right) + \cos\left(\frac{n\pi}{2} + x\right) \right]$$

$$9) y = \sin^2 x$$

$$y = \frac{1}{2} [1 - \cos 2x]$$

$$y = \frac{1}{2} - \frac{1}{2} \cos 2x$$

$$y_n = \frac{1}{2} \left[-2^n \cos\left(2x + \frac{n\pi}{2}\right) \right]$$

$$\left\{ \text{If } y = \cos(ax+b) \text{ then } y_n = a^n \cos\left(ax+b+\frac{n\pi}{2}\right) \right\}$$